

3. Suppose an  $n \times n$  matrix  $A$  is invertible. What can you say about  $\text{Col } A$ ?  
About  $\text{Nul } A$ ?

**Solution:** By the Invertible Matrix Theorem and the equivalence of the statements (a), (e) and (h), we can say that the columns of  $A$  form a linearly independent set and the columns of  $A$  span  $\mathbb{R}^n$ .

By Defn,  $\text{Col } A$  represents the set of all linear combinations of the columns of  $A$ .  $\therefore \text{Col } A = \mathbb{R}^n$ .

$\mathbb{R}^n$  is a subspace of itself.  $\therefore$  The basis of  $\mathbb{R}^n$  are the columns of  $A$ .

By the Invertible Matrix Theorem and the equivalence of the statements (a) and (d), we know that  $Ax = 0$  has only the trivial solution.

$\therefore$  By Defn,  $\text{Nul } A = \{0\}$ .  $\square$